ASHU KUMAR MAURYA

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SUMMARY

Full-Stack Web Developer and Computer Science and Engineering undergraduate with a strong foundation in Data Structures, Algorithms, and Object-Oriented Programming. Skilled in building responsive user interfaces and secure RESTful APIs with JavaScript, React.js, Node.js, Express.js, and MongoDB (MERN Stack). Experience applying Machine Learning to real-time systems, with 300+ solved algorithmic problems and Git-based end-to-end development.

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), HTML5, CSS3
Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, Tailwind CSS, REST APIs
Databases: MongoDB
Tools: Git, GitHub, VS Code, Postman
Core Competencies: Data Structures & Algorithms, OOP, Database Design, RESTful API Design, Machine Learning (Random Forest, XGBoost, Scikit-learn), Version Control
Soft Skills: Problem-Solving, Team Collaboration, Communication, Time Management

PROJECTS

AI-Based Traffic Congestion Prediction System | React.js, Python, Node.js **May 2026 – Sep 2026**

- Developed an intelligent traffic management system using machine learning (Random Forest, XGBoost) to predict congestion and dynamically adjust traffic signal timings.
- Built a full-stack application with a React.js analytics dashboard, a Python/FastAPI ML backend, and a Node.js API/simulation service, backed by SQLite.
- Implemented a safety-critical signal timing engine with conflict prevention and adaptive control logic, reducing vehicle waiting times and improving throughput versus fixed-time signals.

Voice Assistant | HTML5, CSS3, JavaScript **Jan 2025 – Mar 2025**

- Developed a client-side Voice Assistant supporting 15+ real-time voice commands using the Web Speech API and JavaScript.
- Constructed a mobile-first responsive interface across 5+ screen resolutions, reducing visual layout shifts by 100%.
- Streamlined asynchronous event handlers and speech synthesis pipelines to achieve sub-300ms command execution latency.
- Engineered automated fallback validation and speech error handling, ensuring 98% command recognition accuracy across test scenarios.

Agricultural Equipment Rental Platform | React.js, Node.js, Express.js, MongoDB **Jan 2026 – Jul 2026**

- Architected a full-stack rental platform supporting 4+ tractor categories with real-time inventory tracking and date-based scheduling.
- Built a comprehensive owner management portal featuring 10+ REST endpoints to manage equipment inventory, track bookings, and calculate revenue metrics.
- Implemented JWT authentication, 8+ secure route guards, and MongoDB schema validation to eliminate 100% of double-booking scheduling conflicts.

EDUCATION

Ajay Kumar Garg Engineering College, Ghaziabad, Uttar Pradesh **2023 – 2027**
Bachelor of Technology in Computer Science and Engineering *CGPA: 7.0/10*
Relevant Coursework: Data Structures & Algorithms, Web Development, Operating Systems, Computer Architecture

Maa Vindhyavasini Inter College, Paanpi, Sonbhadra **2021 – 2022**
Intermediate (Class XII) *Aggregate: 60.0%*

Maa Vindhyavasini Inter College, Paanpi, Sonbhadra **2019 – 2020**
High School (Class X) *Aggregate: 64.5%*

CERTIFICATIONS

- **Full Stack Web Development** — 6-Week Job-Oriented Training by CodeSquadz & AppSquadz, in collaboration with Ajay Kumar Garg Engineering College, Ghaziabad (Jun 2026 – Jul 2026).

HONORS & ACHIEVEMENTS

- **LeetCode Consistency:** Solved 300+ algorithmic problems while sustaining an active 200-day streak covering arrays, linked lists, trees, and memoization algorithms.
- **Technical Workshop:** Selected for AACAR 9.0 and completed 16 hours of intensive training in Automotive Technology hosted by SAE India Collegiate Club & Bosch JCC.
- **Technical Coursework:** Completed 100+ hours of structured DSA training (College Wallah) and full-stack React.js specialization (Love Babbar).

INTERESTS

Playing Cricket • Exploring Emerging Technologies